Tutorial- 9

1 A retail company wants to predict customer purchase behavior using KNN and evaluate it using K-fold cross validation.

Input features:

- Age
- Annual Income (in thousands)
- shopping frequency (visits per month)

Target variable:

- Purchase (yes=1, No=0)

Dataset.

Customer ID	Age	shopping frequency	Purchase.
1	25	8	Y
2	45	3	N
3	28	9	Y
4	50	2	N
5	30	10	Y
6	48	3	N
7	26	8	Y
8	47	2	N
9	29	9	Y

3- fold cross validation

Cv fold = 3

knn - k = 3

9 ..... 3 fold cv

$9/3 = 3$

$3 \rightarrow$ testing

$6 \rightarrow$ Training.

First testing point (26, 8)

Distance $= \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

$(25,8)\ (26,8) = \sqrt{(26-25)^2+(8-8)^2} = 1$

$(26,8)\ (45,3) = \sqrt{(26-45)^2+(8-3)^2} = \sqrt{361+25} = 19.6$

$(26,8)\ (28,9) = \sqrt{(26-28)^2+(8-9)^2} = 2.23$

$(26,8)\ (50,2) = \sqrt{(26-50)^2+(8-2)^2} = \sqrt{576+36} = 24.7$

$(26,8)\ (30,10) = \sqrt{(26-30)^2+(8-10)^2} = \sqrt{16+4} = 4.4$

$(26,8)\ (48,3) = \sqrt{(26-48)^2+(8-3)^2} = \sqrt{484+25} = \sqrt{50}$
$= 22.5$

Shortest 3 distances 1, 2.23, 4.4

The actual output

$1 \rightarrow$ Yes

$2.23 \rightarrow$ Yes

$4.4 \rightarrow$ Yes.

So output is Yes

Now

Age	S.P (x_2)
47	2

$(47,2)\ (25,8) = \sqrt{(47-25)^2+(2-8)^2} = \sqrt{484+36} = 22.8$

$(47,2)\ (45,3) = \sqrt{(47-45)^2+(2-3)^2} = 2.23$

$(47,2)\ (28,9) = \sqrt{(47-28)^2+(2-9)^2} = \sqrt{361+49} = 20.2$

$(47,2)\ (50,2) = \sqrt{(47-50)^2+(2-2)^2} = 3$

$(47,2)\ (30,10) = \sqrt{(47-30)^2+(2-10)^2} = \sqrt{289+64} = 18.7$

$(47,2)\ (48,3) = \sqrt{(47-48)^2+(2-3)^2} = 1.41$

Shortest distances 1.41, 2.23, 3

$$\left.\begin{array}{l} 1.41 \rightarrow \text{NO} \\ 2.23 \rightarrow \text{NO} \\ 3 \rightarrow \text{NO} \end{array}\right\} \text{NO}$$

→

Age	x_2
29	9

$(29,9)\ (25,8) = \sqrt{(29-25)^2+(9-8)^2} = \sqrt{16+1} = 4.12$

$(29,9)\ (45,3) = \sqrt{(29-45)^2+(9-3)^2} = \sqrt{256+36} = 17.0$

$(29,9)\ (28,9) = \sqrt{(29-28)^2+(9-9)^2} = 1$

$(29,9)\ (50,2) = \sqrt{(29-50)^2+(9-2)^2} = \sqrt{441+49} = 22.1$

$(29,9)\ (30,10) = \sqrt{(29-30)^2+(9-10)^2} = 1.41$

$(29,9)\ (48,3) = \sqrt{(29-48)^2+(9-3)^2} = \sqrt{361+36} = 19.92$

Shortest 3 distances

1, 1.41, 4.12.

$1 \rightarrow Y$
$1.41 \rightarrow Y$
$4.12 \rightarrow Y$
} Yes

We got Yes NO & Yes

The actual output

Customer ID	Age	Frequency	output	predic
7	26	8	Y	Y
8	47	2	N	N
9	29	9	Y	Y

Accuracy $\rightarrow$ 100%.

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